

Sensors For CONSUMER ELECTRONICS

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In this article we discuss the sensors that are used in specific areas of consumer electronic products

In electronics, whether you call it sixth or seventh sense, it is essentially the application of sensors with or without computer programs. Whether it is an invisible computer mouse, a blinkbot, a SixthSense device (invented by Pranav Mistry), a mind-controlled robot, a hand-gesture or a wirelessly-controlled robot, sensors have an important role to play. From smart beverage vending machines to smart home automation systems, from personal assistant robots to smart healthcare systems, from simple computers to digital assistants (like Alexa), from human wearable devices to autonomous vehicles, sensors are part of all these systems. As demand for consumer electronics grows, so does the market for sensors.

Consumer electronics

According to Webopedia, "Consumer electronics refer to any device containing an electronic circuit board that is intended for everyday use by individuals. This encompasses a massive category of electronics that includes

televisions, cameras, digital cameras, PDAs, calculators, VCRs, DVDs, clocks, audio devices, headphones, tablets, smartphones and many other home products."

Consumer electronic products also include radio receivers, mp3 players, video recorders, camcorders, personal computers, video game consoles and automotive electronics, among others.

Although, consumers are rapidly adopting new and emerging products like wearables, voice-activated smart speakers, smart home devices, traditional consumer products such as smartphones, laptops and televisions continue to surpass expectations.

One common thing between most consumer

electronic products is the sensor component. A sensor could be a bare component, a module, active, passive, an input switch, wired or wireless.

Sensors are used extensively in consumer electronics for monitoring, measuring, data logging and control. Common sensors are pressure, proximity, motion, temperature, radio frequency (RF), ultrasonic, flow, level, acoustic, touch and optical.

As technology advances, sensors are becoming much smaller, affordable and are much more power efficient. We are witnessing a growing demand for micro-electro-mechanical systems (MEMS) and nano-electro-mechanical systems (NEMS) in the market. Modern sensor technologies enable manufacturers to use these in portable devices, smartphones, tablets, gaming devices, navigation and home appliances. Sensor technologies are also finding applications in smart home automation, security systems, robotics, automotive and health care, among others.

Major players in the consumer electronics market are Samsung Electronics, LG Electronics, Apple, General Electric, Sony, Huawei, Xiaomi, Bose, Philips, Canon, Haier, Nikon and others. Cities like Shenzhen, Qingdao, Shanghai and Beijing have become important production centres for the industry, attracting many consumer electronic companies across the globe.

Technology trade shows like Consumer Electronics Show (CES), CeBIT and IFA are among the most popular and biggest trade shows in the world. Consumer electronics are the main attractions here.

Sensors in specific applications

Let us look at a few examples where sensors are used in specific areas of consumer electronic products.

Televisions. The most common sensor is the television remote control system. The infrared (IR) sensor in the remote control is used to turn the television on and/or off, change channels, control volume and other settings.

Then, there are sensors in home theatre

Fig. 1: WUW, a type of SixthSense device, during demonstration (Credit: Wikipedia)

